# PANJAB UNIVERSITY, CHANDIGARH

(Estted. under the Panjab University Act VII of 1947- enacted by the Govt. of India)

## FACULTY OF ARTS

## SYLLABI

## *FOR*

## MASTERS IN REMOTE SENSING AND GEOGRAPHIC INFORMATION SYSTEMS (GIS) (SEMESTER SYSTEM)

## EXAMINATIONS, 2012-2013

# PANJAB UNIVERSITY, CHANDIGARH

## OUTLINES OF TESTS, SYLLABI AND COURSES OF READING FOR MASTERS IN REMOTE SENSING AND GEOGRAPHIC INFORMATION SYSTEMS (GIS) (SEMESTER SYSTEM) FOR THE EXAMINATIONS OF 2012 -2013.

## COURSE STRUCTURE

**SEMESTER I** **Max. Marks**

Paper I: Fundamentals of Remote Sensing, Photogrammetry & GPS **200**
- (a) Theory : 100
- (b) Practicals : 60
- (Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks)
- (c) Internal Assessment : 40

Paper II: Fundamentals of Geographic Information Systems (GIS) **200**
- (a) Theory : 100
- (b) Practicals : 60
- (Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks)
- (c) Internal Assessment : 40

Paper III: Image Processing **200**
- (a) Theory : 100
- (b) Practicals : 60
- (Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks)
- (c) Internal Assessment : 40

**SEMESTER II**

Paper IV: Remote Sensing and GIS in Human geography (with special reference to Urban and Regional Planning) **200**
- (a) Theory : 100
- (b) Practicals : 60
- (Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks)
- (c) Internal Assessment : 40

Paper V: Remote Sensing and GIS in Physical Geography (with special reference to environmental management) **200**
- (a) Theory : 100
- (b) Practicals : 60
- (Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks)
- (c) Internal Assessment : 40

Paper VI: Cartography **100**
- (a) Theory : 50
- (b) Practicals : 30
- (Lab. work 15 marks+ Record file 10 marks+ Viva 5 marks)
- (c) Internal Assessment : 20

**Field Report** **100**
- (a) Evaluation of the Field Report : 50
- (b) Presentation : 30
- (c) Viva Voce : 20

**SEMESTER III**

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| Paper VII: | Application of Remote Sensing and GIS in Disaster Management | **200** |
| | (a) Theory : 100 | |
| | (b) Practicals : 60 | |
| | (Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks) | |
| | (c) Internal Assessment : 40 | |
| Paper VIII: | Application of Remote Sensing and GIS in Utility Management | **200** |
| | (a) Theory : 100 | |
| | (b) Practicals : 60 | |
| | (Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks) | |
| | (c) Internal Assessment : 40 | |
| Paper IX: | Application of Remote Sensing and GIS in Resource Management | **200** |
| | (a) Theory : 100 | |
| | (b) Practicals : 60 | |
| | (Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks) | |
| | (c) Internal Assessment : 40 | |

**SEMESTER –IV**

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|---|---|---|
| Paper X: | Research Methodology, Project Formulation and Writing | **100** |
| | (a)Theory : 80 | |
| | (b) Internal Assessment : 20 | |
| | Project Report | **500** |
| | (a) Mid-Term Appraisal of the Project Report : 100 | |
| | (b) Evaluation of the Project Report : 325 | |
| | (c) Presentation &Viva Voce (50+25) : 75 | |
| | **Grand Total** | **2400** |

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**NOTE:**

1. Those candidates who wish to discontinue after successful completion of first two semesters of the Masters course shall be awarded Diploma in Remote Sensing and Geographic Information Systems (GIS).
2. The minimum marks required to pass the examination shall be 45% in each (i) theory paper, (ii) practical work, and (iii) field/project report, and 50% in aggregate.
3. The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department, and one additional faculty member teaching the course.

## SEMESTER - I

**PAPER - I: FUNDAMENTALS OF REMOTE SENSING, PHOTOGRAMMETRY & GPS**
**Marks: 200**

**(a) THEORY** **Marks: 100**
**Time : 3 Hours**

**OBJECTIVES:**

- to introduce the students to the basic principles of remote sensing;
- to indicate the methods of visual and digital interpretation of aerial photographs and satellite imageries;
- to explain the methods of acquiring remote sensing data and their usage; and
- to train the students about elementary photogrammetry and use of GPS.

**COURSE CONTENTS:**

**UNIT – I**

Introduction to remote sensing: Energy and radiation principles, energy-atmosphere interaction, energy-earth surface features interaction, spectral signatures.
Energy Recording Technology: Aerial Photography; Electronic Imaging Devices: Matrix Array, Pushbroom, Wjiskbroom. Remote Sensing Detectors
Resolution of Remote Sensing data: Spatial, spectral, radiometric and temporal.

**UNIT – II**

Type of aerial photographs and satellite imageries: Classification, characteristics and applications.
Sensor Platforms: Ground based, air borne and space borne, earth resources satellite systems: LANDSAT, SPOT, IRS, ERS, IKONOS, QUICKBIRD.

**UNIT – III**

Photogrammetry: Simple geometry of vertical photograph measurement of scale, relief displacement, parallax and height from vertical photographs, Flight Planning.

**UNIT – IV**

GPS-System Description, Historical Background, GPS Segment, Features of GPS Satellite, Principles of Operation, Present Status.
Surveying with GPS-Methods of observation, Absolute Positioning, Relative Positioning, Differential GPS, Kinematic GPS.
GPS Receivers: Navigation Receivers, Surveying and Mapping Receivers, Geodetic Receivers; Computation of coordinates; Application of GPS

**NOTE:**

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit, and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

**(b) PRACTICAL** **Marks: 60**

1. Arrangement of aerial photographs with the help of numbers given on the photograph and preparation of photo index.

2. Interpretation of information given on the margins of the photographs and satellite imageries.
3. Stereo vision test, selection of stereopair, and its orientation and placing under the stereoscope for stereovision.
4. Identification and mapping of features from a single photograph
5. Identification and mapping of features from a stereopair and their verification in the field.
6. Locating principal point, conjugate principal point and flight line on a stereopair.
7. Measurement of scale, relief displacement, height and slope from vertical photographs.
8. Removing error due to relief displacement on vertical photographs by graphical method.
9. A Field based survey on GPS.

**NOTE:** The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

**(c) INTERNAL ASSESSMENT** **Marks: 40**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the university.

## BOOKS RECOMMENDED

**ESSENTIAL READINGS:**


American Society : *Manual of Remote Sensing*, Vol. 1 and II, Falls Church, of Photogrammetry Virginia, 1975.

Barret, E.C. & Curtis, E.F. : *Remote Sensing of Environment*, Second Edition, 1982.

Curran, Paul J. : *Principles of Remote Sensing*, Longman, Hongkong, 1988.

Jensen, J.R. : *Introductory Digital Image Processing: A Remote Sensing Perspective*, Prentice Hall, New Jersey, 1986.

Lillesand, T.M. & Kiefer, : *Remote Sensing and Image Interpretation*, Third R.W. Edition, John Wiley and Sons, 1994.


**FURTHER READINGS:**


Agarwal, C.S. and P.K. Garg : *Text Book on Remote Sensing in Natural Resources Monitoring and Management*, Wheeler Publishing Co, New Delhi, 2000.

Jain, A.K. : *Fundamentals of Digital Image Processing*, Prentice Hall, New Jersey, 1989.

Muralikrishna, I.V. : *Remote Sensing Applications and Geographic Information Systems Recent Trends*, Tata McGraw Hill, New Delhi, 1992.

Fratt, W.K. : *Digital Image Processing*, John Wiley & Sons, 1978.

Rashid, S.K. : *Remote Sensing in Geography*, Manak Publications, Delhi, 1993.

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|---|---|---|
| Rees, W.G. | : | *Physical Principles of Remote Sensing*, Cambridge University Press, Cambridge, 1990. |
| Richards, John, A. | : | *Remote Sensing*: Digital Image Analysis, Springer Verlag, 1986. |
| Swain, Phillip, H. | : | *Remote Sensing*: The quantitative Approach, McGraw Hill International Book Co., 1978. |
| Verbyla, David, L | : | *Satellite Remote Sensing of Natural Resources*, Lewis Publishers, New York, 2005. |


**PEDAGOGY:**

- Students may be taken to field for verification of identifiable features on aerial photographs and satellite imageries;
- Students may be taken to different institutions, such as IIRS, NRSA, State Remote Sensing Centres to acquaint with different equipments, techniques, and their products; and
- Students may be asked to prepare a report using toposheets, aerial photographs and satellite imageries to trace the temporal and spatial changes.

## PAPER II: FUNDAMENTALS OF GEOGRAPHIC INFORMATION SYSTEMS (GIS)

**Marks: 200**

**(a) THEORY** **Marks: 100**

**Time : 3 Hours**

**OBJECTIVES:**

The main objective of this course is to expose the students to fundamental principles of Geographical Information Systems including basic concepts and definitions, methods and techniques.

**COURSE CONTENTS:**

### UNIT – I

Concepts and Definitions:

- Geographic Information Systems (GIS): Definition and developments
- Spatial data base: Points, Lines and Polygons
- Computer components: Hardware and software

### UNIT – II

Functional Elements:

- Data acquisition, input, editing, data manipulation and topology creation
- Data analysis and query
- Symbolisation and product generation

### UNIT – III

Data Management and Structure:

- Data Base Management System (DBMS): Purpose and structure
- Classification: Raster, Vector, quadtree, hierarchical, network and relational

### UNIT - IV

Human and Organizational Issues:

- GIS Applications, GIS users, Hardware and Software questions for organization considering GIS.
- GIS: Decision Support System

**NOTE:**

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit, and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

**(b) PRACTICAL** **Marks: 60**

1. Windows/ NT Environment: Basics, file and directory organisation and management.
2. Familiarization with GIS Software
3. Representation of Real World in GIS: Point, Line and Polygon.
4. Linking Spatial and Attribute Data
5. Spatial data analysis techniques: Buffering, Overlay and Querying.

**NOTE:** The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

**(c) INTERNAL ASSESSMENT** **Marks: 40**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the university.

**BOOKS RECOMMENDED**

**ESSENTIAL READINGS**


| | | |
|---|---|---|
| Aronoff, S. | : | *Geographic Information Systems: A Management Perspective***,** WDL Publications Ottawa,Canada, 1992. |
| ESRI | : | *Understanding GIS*, Environmental Systems Research Institute, U.S.A., 1993. |
| Jefrey S. & John, E. | : | *Geographic Information Systems - An Introduction*, Prentice Hall, New Jersey, USA, 1990. |
| Michael F. Goodchild | : | *Introduction to GIS*, Santa Barbara, California, & Karan K. Kemp (eds.) NGGIA, 1990. |
| Ian Heywood, Sarah Cornelius, Steve Carver | : | An Introduction to Geographical Information Systems Prentice Hall College Div, New Jersey, USA, 2000. |
| Kang-tsung (Karl) Chang | : | Introduction to Geographic Information Systems, 4th Edition, Tata McGraw-Hill, New Delhi, 2009. |


**FURTHER READINGS**


| | | |
|---|---|---|
| Elaine, Rich | : | *Artificial Intelligence*, McGraw Hill Book Co., 1983. |
| Giarratano, J. & Rieley, G. | : | *Expert Systems: Principles and Programming*, PRS Kent Publishing Co., 1985. |
| Laurini, R. & Thomson, D. | : | *Fundamentals of Spatial Information Systems*, Academic Press, London, 1994. |
| Rashid, S.M. | : | *Remote Sensing in Geography*, Manak Publications, Delhi, 1993. |


**PEDAGOGY:**

Taking students for fieldwork, inviting professionals for lectures, taking students to institutes/training centres engaged in imparting professional training and providing research consultancy in business, tourism and area/regional planning should be the main focus.

**Paper III: IMAGE PROCESSING** **Marks: 200**

**(a) THEORY** **Marks: 100**
**Time : 3 Hours**

**OBJECTIVES:**

The course is designed to introduce the student to analytical tools and methods which are currently used in digital image processing as applied to image information for human viewing. Then apply these tools in the laboratory in image restoration, enhancement and compression.

**UNIT – I**

Image processing: definition and types-visual and digital;
Visual image interpretation: Image characteristics, elements of object identification, image interpretation techniques, field verification.

**UNIT - II**

Digital Image Pre-processing: Sources of Errors, Types of error-Radiometric and Geometric, Image Rectification: Geometric Correction, Radiometric Correction, and Noise Removal.

**UNIT - III**

Image Enhancement Techniques: Contrast enhancement-Linear and Non-Linear, Logarithmic and Exponential, Gaussian stretch, Density slicing,
Spatial Filtering: Low frequency and High frequency, Edge enhancement, Band ratioing and band combination, Principal Component Analysis.

**UNIT - IV**

Digital image classification: Classification scheme-Supervised classification (training site selection and statistical information extraction),
Discriminate functions: Maximum Likelihood classifier, Euclidian distance; Unsupervised Classification, Classification accuracy assessment, Error matrix.

**NOTE:**

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

**(b) PRACTICAL** **Marks: 60**

(i) Familiarization with image processing system
(ii) Loading of image data, identification of objects on video display, study of histograms
(iii) Image enhancement techniques: contrast enhancement, band ratioing, edge enhancement, filtering, density slicing
(iv) Image registration: image to map, image to image

(v) Image classification techniques: supervised and unsupervised
(vi) Accuracy analysis
(vii) Ground data collection for training sets for classification of image

**NOTE:** The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

**(c) INTERNAL ASSESSMENT** **Marks: 40**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the university.

## BOOKS RECOMMENDED

### ESSENTIAL READINGS


Burger W., & Burge, M.J. : *Principles of Digital Image Processing,* Springer-Verlag London Limited, 2009.

Gonzalez, R.C. & Woods, R.E. : *Digital Image Processing*, Second edition, Prentice-Hall, Inc., 2002.

Jensen, J.R. : *Introductory Digital Image Processing: A Remote Sensing Perspective*, Prentice Hall, New Jersey, 1986.

Mather, P.M. : *Computer Processing of Remotely-Sensed Images: An Introduction,* Third Edition, John Wiley & Sons, 2008.


### FURTHER READINGS


Lillesand, T.M. & Kiefer, R.W. : *Remote Sensing and Image Interpretation*, Third Edition, John Wiley and Sons, 1994.

Pratt, W.K. : *Digital Image Processing*, Third Edition, John Wiley & Sons, 2001.

Prithvish Nag and M. Kudrat : *Digital Remote Sensing*, Concept Publishing Company, New Delhi, 1998.

Jain, A.K. : *Fundamentals of Digital Image Processing,* Prentice-Hall, Inc., 2002.


### PEDAGOGY:

Students may be asked to prepare a report on image processing techniques using satellite images. Students will be encouraged to ask questions and participate in classroom discussions.

## SEMESTER II

**PAPER IV: REMOTE SENSING & GIS IN HUMAN GEOGRAPHY Marks: 200**
**(with special reference to Urban and Regional Planning)**

**(a) THEORY Marks: 100**
**Time : 3 Hours**

**OBJECTIVES:**
To provide an understanding of:

- concepts and definitions of urban and regional planning so as to pin point the specific areas, where application of Remote Sensing and GIS technologies will increase efficiency and effectiveness of planning;
- data base creation in terms of various attributes, types and hierarchical scales; and
- planning techniques which help in the location of infrastructural facilities for decentralised urban and regional planning.

**COURSE CONTENT:**

**UNIT - I**

Concepts and definitions: Planning - definition and types; role of remote sensing and GIS in regional planning, tourism and business management.

**UNIT - II**

Regional Planning: Creation of data base for regional planning; land use/ land cover mapping; land use suitability evaluation; updating of land use/ land cover maps; resource analysis for district/ block planning; location of amenities and facilities.

**UNIT - III**

Creation of data base for urban planning; updating of urban land use maps; location of facilities and amenities, urban growth monitoring, and urban change detection; route location and site planning, urban development planning; slum identification and improvement.

**UNIT - IV**

Urban and regional mapping, Base map preparation: regional city and intra–city.
Urban and regional plans formulation: Application of remote sensing & GIS techniques, Case Studies.
Urban analysis: Case studies pertaining to urban growth monitoring, change detection and slum development,

**NOTE:**

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).
2 A total of eight questions will be set out of the whole syllabus, at least 2 from each unit and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

**(b) PRACTICAL** **Marks: 60**

(i) Image/photo interpretation: recognition and detection of objects, preparation of land use maps.
(ii) Digital image processing techniques in preparation of land use and urban land use maps.
(iii) Preparation of base map and its updating.
(iv) Urban growth monitoring and change detection.
(v) Slum identification and mapping.

**NOTE:** The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

**(c) INTERNAL ASSESSMENT** **Marks: 40**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the university.

**BOOKS RECOMMENDED**

**ESSENTIAL READINGS**


Avery, T.V. : *Interpretation of Aerial Photography,* Burgess Publishing Co., Minneapolis, 1962.

Barret, E.C. & Curtis, E.F. : 'The Built Environment' in Remote Sensing of Environment, Second edition, pp. 293-306, 1982.

Gautam, N.C. : *Urban Landuse Study through Aerial Photo Interpretation Techniques*, Pink Publishing House, Mathura, 1970.

Muralikrishna, I.V. : *Remote Sensing Applications and Geographic Information Systems - Recent Trends*, Tata McGraw Hill, New Delhi, 1992.

Nag, Prithvish : Thematic Cartography and Remote Sensing, Concept, New Delhi, 1992.

Richardson, B.F(ed) : Introduction to Remote Sensing of the Environment, Kendall/Hunt, Dubuque, Iowa, 1978.

Ian Heywood, Sarah Cornelius, Steve Carver : An Introduction to Geographical Information Systems Prentice Hall College Div, New Jersey, USA, 2000.

Kang-tsung (Karl) Chang : Introduction to Geographic Information Systems, 4th Edition, Tata McGraw-Hill, New Delhi, 2009.


**FURTHER READINGS**


Sundaram, K.V. : Urban and Regional Planning in India, Concept, New Delhi, 1977.

Taylor, John,L. : Urban Planning Practice in Developing Countries, Williams, David C Pergamon Press, 1981.


**PEDAGOGY:**
Taking students for fieldwork, inviting professionals for lectures, taking students to institutes/training centres engaged in imparting professional training and providing research consultancy in business, tourism and area/regional planning should be the main focus.

**PAPER V: REMOTE SENSING AND GIS IN PHYSICAL GEOGRAPHY Marks: 200**
**(with special reference to Environmental Management)**

**(a) THEORY Marks: 100**
**Time : 3 Hours**

**OBJECTIVES:**

- to train the students to detect, identify and recognise the various facets of different landforms using remote sensing and creating a data base in GIS mode;
- to educate students about morphogenetic analysis using photogrammetry & GIS; and
- to indicate the use of remote sensing and GIS in assessment and management of environmental problems.

**COURSE CONTENT:**

**UNIT - I**

Concepts and definitions:

- Geomorphic Processes, Terrain Elements, Principles of Geomorphological image interpretation, Identification and Reognition of landform
- Qualitative and quantitative study of relief.

**UNIT - II**

Remote Sensing and GIS in Geomorphology:

- Manual and digital methods of morphometric analysis of a drainage basin.
- Use of remote sensing in understanding the relationships between structure and process of landforms, climate and landforms structure and drainage pattern, landforms and surface material, landforms and landuse, landforms and human activities.
- Classification of landforms using remote sensing and GIS

**UNIT - III**

Remote Sensing and GIS in climatology, hydrology and environmental geomorphology

- Use of remote sensing and GIS in recording the meteorological data and weather forecasting.
- Use of remote sensing and GIS in hydrological studies, especially mapping of water resources and water related problems.
- Use of Remote Sensing & GIS in impact assessment of human activities on the geomorphological processes and the landscape

**UNIT – IV**

- Use of Remote Sensing and GIS in assessment of environmental problems such as deforestation, water logging, desertification, water, pollution etc.

**NOTE:**

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number 1.

**(b) PRACTICAL** **Marks: 60**

(i) Identification and mapping of relief, drainage pattern, water bodies and surface material from topographical maps
(ii) Identification and mapping of relief, drainage pattern, water bodies and surface material from aerial photographs and satellite imageries.
(iii) Identification and mapping of terrain units from large scale aerial photographs
(iv) Measurement and mapping of slopes from topographical maps
(v) Rough estimation and mapping of slopes from the aerial photographs
(vi) Morphometric analysis of drainage basins from topographical maps
(vii) Analysis of certain selective aspects of the morphometry of the drainage basins from the aerial photographs and satellite imageries
(viii) Empirical quantitative classification of terrain from topographical maps
(ix) Genetic and functional classification of terrain from the aerial photographs and satellite imageries
(x) Identification and mapping of impact of human activities on the geomorphological processes and landscape using remote sensing and GIS techniques

**NOTE:** The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

**(c) INTERNAL ASSESSMENT** **Marks: 40**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the university.

**BOOKS RECOMMENDED**

**ESSENTIAL READINGS**


Agarwal, C.S. and P.K. Garg : *Text Book on Remote Sensing in Natural Resources Monitoring and Management*, Wheeler Publishing Co, New Delhi, 2000.

Druary, S.A. : *Image Interpretation in Geology*, Allen and Unwin Ltd., London, 1987.

Siegal, B.S. & Gillespie, A.R : *Remote Sensing in Geology*, John Wiley Publications, 1986.

Townsend, J.R.G. : *Terrain Analysis and Remote Sensing*, George Allen and Unwin, 1981.

Verstappen, H. : *Remote Sensing in Geomorphology*, Elseiver Scientific Publications, Netherlands, 1977.

Verbyla, David, L : *Satellite Remote Sensing of Natural Resources*, Lewis Publishers, New York, 2005.


**FURTHER READINGS**


Greedry, Alan, F. : Application of Remote Sensing with Special Reference to Geosciences, Gregory Geo-science, 1974.

Nefedov, K.E. : Deciphering of Groundwater from Aerial Photographs, Amerind Publications, Delhi, 1972.

Papova, T.A. and Pandey, S.N. : Principles and Applications of Photogeology, Wiley Eastern Publishers, 1987.

Smith, Willam, L. : Remote Sensing Applications for Mineral Exploration, Dawden Hutchintons and Ross Inc.,1977.

**PEDAGOGY:**

Students may be taken to institutions such as IIRS, NRSA, State Remote Sensing Cetres to acquaint them with equipments, techniques and their products. Students may be asked to prepare a report on landform using topographical sheets, aerial photographs and satellite images. Students may acquaint with the satellite imageries of various kinds of environmental problems.

**PAPER VI: CARTOGRAPHY** **Marks: 100**

**(a) THEORY** **Marks : 50**
**Time : 3 Hours**

**OBJECTIVES:**

The course is designed to emphasise the relevance and scope of cartography in GIS. It shall highlight the value of this vital field in the current and highly versatile field of GIS.

**UNIT - I**

- Cartography: Definition, Scope and Historical Perspective.
- Cartography and GIS
- Traditional vs Computer (Digital ) Cartography

**UNIT - II**

- Types of Data: Spatial data and Attribute data
  Spatial Data: Point, line, area
  Attribute Data: Data Analysis and Classification
- Map Types: Quantitative and Qualitative

**UNIT - III**

- Map Design: Principle of Symbolization , Colour and Colour Schemes, Map Lettering, Map Layout, Legend Presentation
- Topographic Maps: General Introduction, Types and Classification

**UNIT - IV**

- Techniques of Relief Representation and Interpretation: Understanding Hill Shading, Hachures and Contours
- Profiles ( longitudinal and Transverse) and Slope Analysis

**NOTE**:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, atleast 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks (36 marks). These will be in addition to the compulsory question at serial number I.

**(b) PRACTICAL** **MARKS: 30**

**Exercises:**

1. Classification of topographical sheets on various scales as per Old Map Series of India
2. Classification of topographical sheets on various scales as per New Map Series of India
3. Interpretation of Physical and Cultural aspects of an area using SOI topographical sheet on 1:50,000 scale.
4. Creation of Digital Elevation Model (DEM) using topographical sheets.
5. Generation of Slope Map using DEM
6. Generation of Aspect Map using DEM
7. Generation Hill Shading Map using DEM
8. Profile generation using Topographical Sheets: Longitudinal Profile, Transverse Profile
9. Profile generation using DEM: Longitudinal Profile, Transverse Profile and Surface Profile
10. Generation of Thematic Maps: Dot Map, Choropleth Map, Proportionate Circles, Isopleth Map

**Software Used:** Arc GIS and ERDAS IMAGINE

**NOTE:** The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

**(c) INTERNAL ASSESSMENT** **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the university.

**BOOKS RECOMMENDED**

**ESSENTIAL READINGS**


| | | |
|---|---|---|
| Cromley, R.G. | : | *Digital Cartography* ,Prentice- Hall of India, New Delhi, 1992. |
| Crampton, W.C. | : | *Mapping :A Critical Introduction to Cartography and GIS,* John Willy & Sons, New York, 2010 |
| Dent, B.D. | : | Cartography-*Thematic Map Design*, WBC McGraw–Hill, Boston, 1999. |
| John Krygier and Denis Wood. | : | Making Maps: *A visual guide to map design for GIS,* Guilford Publications, New York, 2005. |
| Keates, J.S. | : | *Cartographic Design and Production*, Longman, London, 1998. |
| Misra,R.P.and Ramesh, A. | : | *Fundamental of Cartography*, Concept Publishing Company, New Delhi, 1989. |
| Monkhouse, F.J. | : | *Maps and Diagrams*, Methuen and Co., London, 1994. |
| Nag, P. | : | Thematic Cartography and Remote Sensing, Concept Publishing Co., New Delhi ,1992. |

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|---|---|---|
| Rampal, K.K. | : | *Mapping and Compilation*, Concept Publishing Co., New Delhi, 1993. |
| Raisz, Erwin | : | *Principles of Cartography*, McGraw Hill, New York, 1962. |
| Robinson, A.H. and others | : | *Elements of Cartography*, John Willy & Sons, New York, 6th edition, 1992. |
| Singh, R.L.and Singh R.P.B | : | *Elements of Practical Geography,* Kalyani Publishers , New Delhi, Reprint 2002. |


**FURTHER READINGS**


| | | |
|---|---|---|
| Birch, T. | : | *Maps- Topographical and Statistical*, Clarendon Press, Oxford, 1949. |
| Brown, L.A. | : | *The Story of Maps,* Cressit Press, London, 1951. |
| Harley J.B. and David Woodward | : | *The History of Cartography* Volume 2, Book 2, The University of Chicago Press, 1995. |
| Ramamurthy, K. | : | *Maps Interpretation: India Landscapes Through Survey of India*, Topographic Maps, R.K. Mutt Road, Madras, 1982. |
| Wood Clifford H. and Keller C. | : | *Cartographic Design- Theoretical and Practical Perspectives*, John Wiley & Sons, 1996. |


**PEDAGOGY:**

Students will be encouraged to ask questions and participate in classroom discussions. Assignments, term papers and group discussions will be the additional methods to make teaching more focused and interesting.

**FIELD REPORT** **Marks: 100**

Distribution of Marks:

a) Evaluation of the Field Report : 50
b) Presentation & Viva Voce (25+25) : 50

**NOTE**:

The field report shall be prepared by the student on the basis of work done or experiences gained on visit(s) to Remote Sensing Centre(s)/Institute(s) in India. The student will submit three copies of his/her work ten days before the commencement of the theory examination of the said semester. Three examiners, consisting of supervisor, chairperson of the department, and one more faculty member teaching the course, shall evaluate the field report independently and student shall be awarded the mean score of the three evaluations. The evaluation process shall have two-components:(i) evaluation of field report, (ii) presentation of data analysis and main findings before the board of examiners and viva voce.

## SEMESTER III

**PAPER VII: APPLICATION OF REMOTE SENSING AND GIS IN DISASTER MANAGEMENT** **Marks: 200**

**(A) THEORY** **Marks: 100**
**Time : 3 Hours**

**OBJECTIVES:**
Main objectives of this course are to introduce the students with the nature, causes and effects of disasters and applications of GIS and Remote Sensing in mapping, mitigation and monitoring of hazards and disasters.

### Unit I

i. Fundamental concepts of hazards and disasters
ii. Natural and human induced disasters: classification, types
iii. Historical perspective of disasters in India

### Unit II

iv. Disaster management: concepts relating to the pre-disaster phase, emergency phase, post-disaster phase
v. Disaster management mechanism in India: agencies, plans and policies
vi. Role of remote sensing and GIS in disaster management

### Unit III

vii. Remote sensing and GIS application in geological hazards: Landslides and Earthquakes.
viii. Remote sensing and GIS application in Hydro meteorological Hazards: Floods, Cyclones, Drought.

### Unit IV

ix. Remote sensing and GIS application in Environmental hazards: Deforestation, Land Degradation and desertification, Forest fires
x. Case Studies: Earthquakes in India, Drought in India, Floods in Indo-Gangetic plains, Landslides in Himalayan region,

**Essential Readings**


Greg Bankoff , Georg Frerks, : Dorothea Hilhorst — *Mapping Vulnerability: Disasters, Development and People*, Earthscan Publications Ltd., 2004.

Monmonier mark, : *Cartographies of Danger: Mapping Hazards in America*, University of Chicago Press, 1997.

Roy, P.S.; Van Westen, C.J.; Jha, V.K.; Lakhera, R.C. and Champati Ray, P.K., : *Natural Disaster and their Mitigation: Remote Sensing and Geographical Information System Perspectives*, IIRS, DehraDun, Govt. of India, India, 2000.

**FURTHER READINGS**


Agarwal, C.S. and P.K. Garg : *Text Book on Remote Sensing in Natural Resources Monitoring and Management*, Wheeler Publishing Co, New Delhi, 2000.

Burrough, P.A. & Macdoneli, R.A.: *Principles of Geographic Information Systems,* Oxford University Press, 2000.

Demers, M.N. : *Fundamentals of geographic Information Systems*, 2nd Edition, John Wiley and sons, 2000.

Giarratano, J. & Rieley, G. : *Expert Systems: Principles and Programming*, PRS Kent Publishing Company, 1985.

Muralikrishna, I.V. : *Remote Sensing Applications and Geographic Information Systems Recent Trends*, Tata McGraw Hill, New Delhi, 1992.

Rashid, S.K. : *Remote Sensing in Geography*, Manak Publications, Delhi, 1993.

Rees, W.G. : *Physical Principles of Remote Sensing*, Cambridge University Press, Cambridge, 1990.

Swain, Phillip, H. : *Remote Sensing*: The quantitative Approach, McGraw Hill International Book Co., 1978.

Verbyla, David, L : *Satellite Remote Sensing of Natural Resources*, Lewis Publishers New York, 2005.


**NOTE:**

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt all 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit.
3. The candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

**(B) PRACTICAL** **Marks: 60**

1. Earthquake hazard mapping using satellite images and ancillary data.
2. Flood prone area mapping using satellite images and ancillary data.
3. Landslide mapping and risk evaluation.
4. Multiple hazard mapping using satellite images, ancillary data and modeling risk in GIS.

**(C) INTERNAL ASSESSMENT** **Marks: 40**

Internal assessment shall be based on class test/s and assignment/s etc., as prescribed or conducted by the teacher/s concerned.

**PEDAGOGY:**

- Students may be asked to prepare a report on zonation of hazards and disasters using topographical sheets, aerial photographs and satellite images.
- Students may acquaint with the satellite imageries of various kinds of disasters.

**PAPER VIII: APPLICATION OF REMOTE SENSING AND GIS IN UTILITY MANAGEMENT** **Marks: 200**

**(A) THEORY** **Marks: 100**
**Time : 3 Hours**

**OBJECTIVES:**

The major objective of this paper is to expose the students the application of remote sensing and GIS in managing utility services. It further aims at training them to take quick, correct and reliable locational decisions.

## Unit I

Utility, Description of all essential services and utilities, Database development and Data Acquisition, Acquiring and integrating geo-spatial data, Spatial Data Bases

## Unit II

Spatial Data Manipulation and Analysis, Geospatial system analysis and design, Geospatial technology project, management, Query Processor and Visualization, Applications and Problem solving with GIS
Electricity, Gas, Water supply, Sewerage system
Solid waste disposal, Telecommunication, Public health and safety, Crime analysis

## Unit III

Modelling in utility applications, Infrastructure aims and objectives, Environmental law and regulations governing infrastructure utilities, Modern infrastructure tools

## Unit IV

Case studies Part I: relating to Electricity, Gas, Water supply, Sewerage system
Case studies Part II: Solid waste disposal, Telecommunication, Public health and safety, Crime analysis

**REFERENCES**


1. Harries K (1999) *Mapping Crime: Principle and Practice*. Washington, DC: Crime Mapping Research Center, Department of Justice, U.S.A.
2. Hodder I, Orton C (1979) *Spatial Analysis in Archaeology.* Cambridge: Cambridge University Press .
3. Escritt, L. B. (1972). *Water Supply and Building Sanitation*, 4th Ed., Mac Donald and Evans Limited, 1972
4. Hammer, Mark J. (1986). *Water and Waste Water Technology*, 2nd Ed., John Willey and Sons Inc., 1986.


**NOTE:**

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit, and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

**(B) PRACTICAL** **Marks: 60**

Site Suitability mapping and locating and relocating of infrastructure facilities related to:-

-Electricity and Gas,

-Water supply and Sewerage system,

-Telecommunications and Transport

-Site suitable for solid waste disposal, and

-Accessibility mapping of health care services.

**NOTE**: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department, and one additional faculty member teaching the course.

**(C) INTERNAL ASSESSMENT** **Marks: 40**

Internal assessment shall be based on class test/s and assignment/s etc., as prescribed or conducted by the teacher/s concerned.

**Paper IX: APPLICATION OF REMOTE SENSING AND GIS IN RESOURCE MANAGEMENT** **Marks: 200**

**(A) THEORY** **Marks: 100**
**Time : 3 Hours**

**OBJECTIVES:**

It aims at training the students to use remote sensing and GIS in management of resources. It will expose them to plan things better and utilize resources most efficiently.

**Unit I**

Concepts in Resources: Resources classification systems, natural and cultural resources, renewable and non-renewable resources.
Resource Conservation: Remote sensing based Land use- Land cover mapping for resource monitoring and management Sustainable development of natural resources.

**Unit II**

Land Resources: Introduction to agriculture and soil resources, Crop inventory, land evaluation, land degradation and soil conservation; Mineral resources and impacts of mining.
Bio-Resources: Remote sensing application in agriculture, forest resources and wildlife habitat assessment. Mapping of forest density and type, issues in forest management.

**Unit III**

Water Resources: Remote sensing application in surface and sub surface water resources evaluation, watershed characterization, water resources management and development, water quality and pollution issues.
Energy Resources: Coal, oil and nuclear energy, non-conventional energy resources, future potential and requirement of energy resources. Geo-information in energy resources management.

**Unit IV**

Geo-information Models in Resource Management: Land resources development, prioritization and modeling, Forest resources modeling, Agriculture production potential modeling.

**NOTE:**

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit, and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

**REFERENCES**:


1. Simmons, T.G. (1974). Ecology of Natural Resources, Publ. Edward Arnold, London.
2. Chouhan, T.S. and Joshi, K.N. (1992). *Remote Sensing for Natural Resources Management*, Universal Scientific Publ. Jaipur.
3. Davidson, Donald A. (1998). *Soils and Land use Planning*, Longman, London.
4. Odum, E.P. (1964). *Fundamental of Ecology*, V.B. Saumders and Co.
5. Miller, R. W. and Donahue, R. L. (1990): *Soils*, Prentice-Hall of India.
6. Lillesand, Thomas M. & Kiefer Ralph (2000): *Remote Sensing and Image Interpretation*, John Wiley.
7. Colwell, Robert W. *Monitoring of Earth Resources from Aircraft and Spacecraft*, NASA, Washington D.C.
8. Ester, J. and Senger, L.W. *Remote Sensing Techniques for Environment Analysis*, Tayler and Francis Inc.


**(B) PRACTICAL** **Marks: 60**

1. LU-LC mapping at level I and Level II using 1:50,000 scale satellite image.
2. Forest Types Mapping using satellite images.
3. Delineating on satellite image various surface water resources and identify potential sites for WR conservation.
4. Delineation of surface mining zones and impact on land resources of the area using satellite image
5. NDVI and density slicing of digital satellite data for forest density classification.
6. Supervised classification for mapping agriculture and forest resources.
7. Applying geo-information models in resource management and development plan using raster GIS.
8. Soil erosion modeling using geo-information approach
9. Natural resource mapping and change detection study using temporal satellite data

**NOTE**: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department, and one additional faculty member teaching the course.

**(C) INTERNAL ASSESSMENT** **Marks: 40**

Internal assessment shall be based on class test/s and assignment/s etc., as prescribed or conducted by the teacher/s concerned.

## SEMESTER IV

**PAPER X: RESEARCH METHODOLOGY, PROJECT FORMULATION AND WRITING**

**Marks: 100**
(Terminal exams: 80 marks)
(Internal Assessment: 20 marks)
**Time : 3 Hours**

**OBJECTIVES:**

The aim of this paper is to train the students about the various aspects of research writing so that they are in a position to write research reports independently.

### Unit I

Problems of Geographical Information research: Identification of problems of regional and Local level, geographic data sources and nature of data to be used.
Hypotheses and Models
Formulation of research schemes

### Unit II

Preparation of research projects and writing of reports: Preparation of field reports, spatial data, classification and sampling problems. Need for sampling, types of sampling, sample size, sampling area.
Project Definition, Importance of Projects and Project Management, Project Management context. Basics of project management, Project formulation, Time management, Budget estimates, Cost-benefit calculation techniques

### Unit III

Project bidding, Project plan, Task definition, Project resource, Scheduling, Line management, Project team.
Managing the Projects Activities, Project administrator, Classification of projects. Product management, Problems and opportunities in Projects.

### Unit IV

Tools & Methods: Project Communications and Presentation, Project Management Software, Project Administration.
Evolution, revolution, & termination of Project, Project change, End of projects, Project report preparation.

**NOTE:**

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).

2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit; and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

3. Mid-term appraisal shall include at least two presentations by the students before final submission of his/her project report. First presentation will be done after the collection of data/information and second one after the mapping and interpretation of the data/information. These presentations shall be assessed by a board of examiners consisting of Chairperson of the Department, faculty member teaching the course and one more senior faculty member.

**REFERENCES**:

1. W.E. Huxold & A.G. Lerinsons Aronoft. S. (1989). *Managing Geographic Information Projects*.

2. Earickson, R., and Harlin, J. (1994). *Geographic Measurement & Quantitative Analysis*, Macmillan, New York

3. Bennet P. Lientz & Kathryn P. (1995). *Project Management for the 21st Century*, Academic Press, California

**PROJECT REPORT** **Marks**: **500**

| | | |
|---|---|---|
| (a) Mid-Term Appraisal of the Project Report | : | 100 |
| (b) Evaluation of the Project Report | : | 325 |
| (c) Presentation &Viva Voce (50+25) | : | 75 |

**NOTE**:

(i) Mid-term appraisal of the project report of each student shall be done by the board of the four examiners consisting of the supervisor, chairman, and the two senior faculty members of the department.

(ii) Mid-term appraisal of the project report shall have two-components: (i) presentation on project proposal including the research objectives, methodology and the chapter scheme, after the three weeks of assigning the project title, (ii) presentation of data analysis and tentative findings, after the six weeks of assigning the project title.

(iii) In case of project report, the Board of Control shall assign a candidate to a faculty member/s for supervision of his/her report on an approved topic. The student shall prepare three copies of his/her project report and submit in the concerned department. The report shall have to be submitted ten days before the commencement of the theory examination of the said semester. The project report shall incorporate the findings arrived at on the basis of the data/information collected from the field and processed, analyzed and mapped in the GIS lab.

For the evaluation of project report, three examiners, consisting of supervisor, chairperson of the department, and one more faculty member teaching the course, shall evaluate the field report, submitted by the student, independently and student shall be awarded the mean score of the three evaluations. The evaluation process shall have the two-components: (i) evaluation of report, (ii) presentation of data analysis and main findings before the board of examiners and viva voce.

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